Geometry of a Mental Task: Topological Data Analysis for EEG Processing

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Abstract

Electroencephalography (EEG) is currently the most widely used non-invasive method for detecting cognitive activity. Being an inherently high-dimensional data generator, a sufficiently robust feature extractor is needed to capture as much information as possible. Topological Data Analysis (TDA) serves as a tool for capturing higher-order relationships between points, as well as their joint geometry. In this paper, we ask whether the functional connectivity of EEG recordings carries sufficient information to discriminate cognitive states. The aim of this research lies in topology's contribution to the *representation* of the data, not in the classifier itself. Using a conventional SVM, the TDA pipeline classified EEG recordings from 36 subjects as either rest or mental arithmetic with an accuracy of 0.89 [0.79-0.97], matching a band-power baseline tuned identically (0.90). This result shows that the geometry of coordinated cortical activity carries sufficient information to discriminate distinct cognitive states, matching representations that use electrode identity.

1 Introduction

Electroencephalography (EEG) is a widely used, non-invasive method that records the electrical activity in the brain; it yields a comprehensive image of brain dynamics over time. Exploiting this technology is key to advancing research towards reliable brain-computer interfaces (BCI), while avoiding the inherent risks of inserting hardware inside the skull [1]. In an attempt to handle the complexity of the brain, topological data analysis (TDA) applied to EEG analysis has grown in medical diagnosis for the study and detection of neurological conditions like epilepsy/seizures [2] and schizophrenia [3].

However, in recent studies [4], baseline-free evaluation has been one of the main weaknesses in EEG-TDA reports, where results are shown without comparing them to what a conventional feature set achieves on the same recordings. Additionally, most research papers are aimed at data scientists, neurobiologists, and engineers who are more focused on the computational and practical side of TDA implementation for brain data analysis; making the mathematical explanation rushed, too general, or not intuitive. We offer an in-depth explanation of how the idea comes to be, starting from easy-to-define concepts like vector spaces and groups. The target audience of this paper is mathematicians, engineers, and scientists who are curious about this tool's mathematical foundations, accompanied by a practical result on an EEG dataset. Understanding how this computational engine works allows one to better implement it by understanding its scope and limitations. Additionally, having a tool like TDA at our disposal offers direct interpretability of the data features that the model classifies, as opposed to abstract, unexplainable features that models tend to infer.

In Section 2, we will go over a relatively self-contained build-up of the mathematical apparatus that enables us to compute the topology of the data. Section 3 covers the dataset, and Section 4 describes the pipeline and the metrics used to determine the relevance of the obtained results. Section 5 states these results, and in Section 6, we discuss their relevance, how the obtained accuracies compare, and the limitations of our pipeline before concluding in Section 7.

2 Background: Algebraic Topology

In this section, we will go over the theory behind TDA, which enables us to study the topology of the data. This is meant to be a relatively self-contained handbook of the underlying structure and implementation of topological data analysis. Readers familiar with simplicial homology may skip to Section 2.3; those familiar with persistent homology may skip directly to Section 3.

2.1 Definitions and build-up

We will build the mathematical framework, as well as the geometric intuition towards *simplicial homology* (which will help us analyze the shape of the data) that can be computed from the relative distances between data points. That being said, we will keep definitions concise with the specific parameters upon which the code is built. This is why we will take definitions not broadly, but as specific instances directed at the workings of the software. As per standard algebraic topology practice, we will first develop singular homology before forking into the simplicial case that the software computes. For more detailed definitions, refer to Hatcher's *Algebraic Topology* [5].

Definition 1. A *group* $(G, *)$ is a set defined under an operation that satisfies the following properties:

- $a, b \in G \implies a * b \in G$ (closure)
- $\forall a, b, c \in G,\ (a * b) * c = a * (b * c)$ (associativity)
- $\exists\, e \in G \ni \forall a \in G,\ \ a * e = a = e * a$ (identity)
- $\forall a \in G,\ \exists\, a^{-1} \ \ni\ a * a^{-1} = e = a^{-1} * a$ (inverse)

Definition 2. A group $(G, *)$ is said to be *abelian* if their elements commute under the given operation. This is, it satisfies:

- $a, b \in G \implies a * b = b * a$ (commutative property)

Definition 3. A *free* abelian group is an abelian group that possesses a basis.

Definition 4. For $n \geq 0$, the *standard n-simplex* Δ^n is the convex hull of the standard basis $\{e_0, e_1, ..., e_n\} \subset \mathbb{R}^{n+1}$:

$$\Delta^n = \{\textstyle\sum t_i e_i : \sum t_i = 1, t_i \geq 0\}$$

It is useful to think of the standard n-simplex as the n-dimensional generalization of a triangle, which is an instance at $n = 2$.

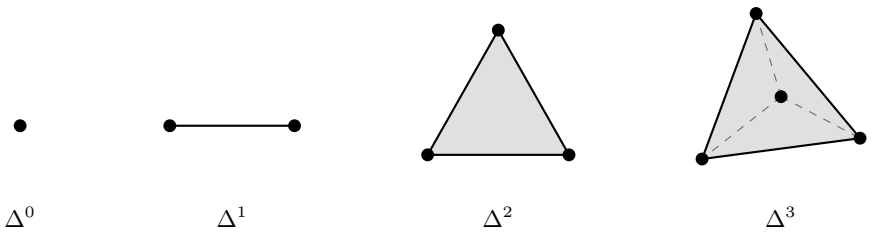


Figure 1: The standard n-simplices for $n = 0, 1, 2, 3$: a point, an edge, a triangle and a tetrahedron. Each Δ^n has $n + 1$ vertices.

More generally, an n-simplex is any affine image of Δ^n under an injective affine map.

Definition 5. A *simplicial complex* K is a collection of simplices of various dimensions that satisfies the following conditions:

- Any face of a simplex in K is itself in K.
- Any two simplices of K intersect either in a common face or not at all (no partial overlaps).

Concretely, K is determined by its vertex set V together with the collections of subsets $\tau \subseteq V$ that are faces. This combinatorial data is what is stored and used computationally.

Definition 6. Let X be any topological space. A *singular n-simplex* in X is a continuous map $\sigma : \Delta^n \to X$. We denote $\mathrm{Sin}_n(X)$ the set of all n-simplices in X.

Note that n-simplices are related by face inclusions $\delta^i : \Delta^{n-1} \to \Delta^n$ for $0 \leq i \leq n$ that satisfy $\delta^i \delta^j = \delta^{j+1} \delta^i \, (i \leq j)$.

Face inclusions send the vertices $e_0, e_1, ..., e_{n-1}$ in order to $e_0, ..., \hat{e}_i, ..., e_n$ - i.e. the order-preserving injection that omits e_i.

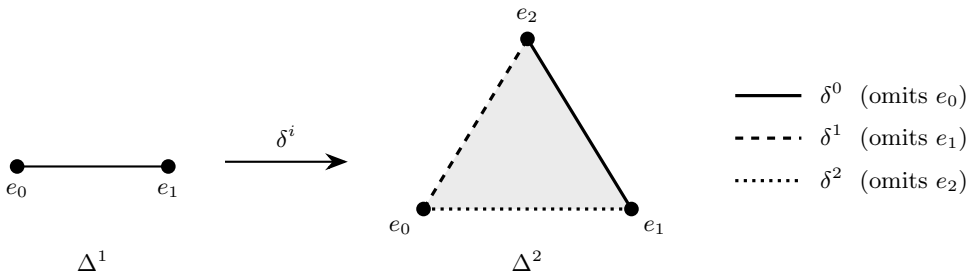


Figure 2: The face inclusions $\delta^i \colon \Delta^1 \to \Delta^2$. Each δ^i is the order-preserving injection onto the edge of Δ^2 obtained by omitting the vertex e_i.

These produce maps $d_i : \mathrm{Sin}_n(X) \to \mathrm{Sin}_{n-1}(X)$ that send $\sigma \mapsto \sigma \circ \delta^i$ which corresponds to the "i-th face" of σ.

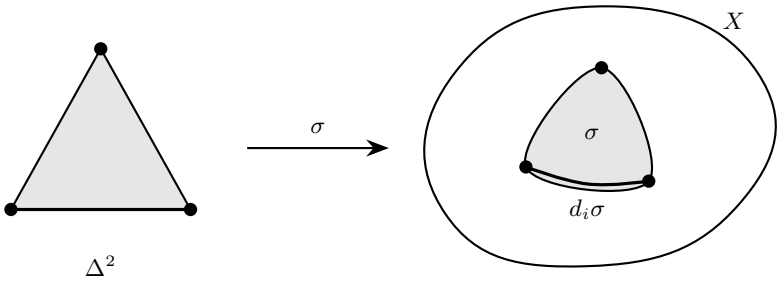


Figure 3: A singular 2-simplex $\sigma \colon \Delta^2 \to X$. Precomposing with a face inclusion, $d_i\sigma = \sigma \circ \delta^i$, gives the i-th face of σ: a singular 1-simplex in X (highlighted).

It is a good place to state that the difference in the two main types of homology resides in *simplicial homology* requiring the topological space X to be equipped with a simplicial complex structure, or a "triangulable" space. However, when X is homeomorphic to a geometric realization of a simplicial complex, then the simplicial homology of X agrees with its *singular homology* (the one computed with singular simplices), meaning that they compute the same topological invariants [Hatcher, *Algebraic Topology*, Theorem 2.27]. In particular, this is true when X is itself a simplicial complex. In our setting, the ambient space is never given explicitly; we only get a sample. Building a simplicial complex from that sample yields an object whose homology reduces to rank computations on finite matrices, which is the method that the software implements.

Definition 7. An *n-chain* is a finite linear combination of simplices,

$$\sum_{i=1}^{k} z_i \sigma_i, \quad z_i \in \mathbb{Z}, \quad \sigma_i \in \mathrm{Sin}_n(X)$$

Definition 8. The abelian group $C_n(X)$ of *singular n-chains* in X is the free abelian group generated by n-simplices

$$C_n(X) = \mathbb{Z}\,\mathrm{Sin}_n(X)$$

Now we can define the *boundary operator*

$$\partial : \mathrm{Sin}_n(X) \to C_{n-1}(X)$$

by

$$\partial\sigma = \sum_{i=0}^{n} (-1)^i d_i \sigma$$

This extends to a homomorphism $\partial_n : C_n(X) \to C_{n-1}(X)$ by additivity, as can be uniquely done by any function out of the basis. This is because of the universal property of the free abelian group: any function out of a basis extends uniquely to a homomorphism.

Concretely, if $c = \sum_j c_j \sigma_j$ then $\partial_n c = \sum_j c_j \partial_n \sigma_j$

When X is a simplicial complex K, this boundary operator restricts to the combinatorial faces of K. For an ordered face $\tau = [v_0, \ldots, v_n]$,

$$\partial_n \tau = \sum_{i=0}^{n} (-1)^i [v_0, \ldots, \hat{v}_i, \ldots, v_n]$$

where $\hat{v}_i$ denotes omission.

Let $C_n(K) := \mathbb{Z}\{n\text{-faces of } K\}$ denote the free abelian group on the (ordered) n-faces of K; these are the simplicial chains, and the boundary above restricts to them. This yields a family of free abelian groups that are connected through homomorphisms:

$$C_{n+1}(K) \xrightarrow{\partial_{n+1}} C_n(K) \xrightarrow{\partial_n} C_{n-1}(K)$$

The family forms a *chain complex* (C_i, ∂_i) for $i \geq 0$ with $C_{-1} = 0$ and $\partial_0 = 0$. Here, each n-face of K is a generator of the group.

Definition 9. An *n-cycle* in K is an n-chain c such that $\partial_n c = 0$.

Theorem 1. Any boundary is a cycle, i.e.

$$\partial_n \circ \partial_{n+1} = 0$$

We will skip the proof of this theorem, but it follows from the simplicial identity $\delta^i \delta^j = \delta^{j+1} \delta^i \, (i \leq j)$ [Hatcher, *Algebraic Topology*].

However, we will see the triangle example to build on intuition. Let $\sigma = [v_0, v_1, v_2]$ be a 2-simplex. If we take the boundary

$$\begin{aligned} \partial_2 \sigma &= d_0 \sigma - d_1 \sigma + d_2 \sigma \\ &= \sigma \circ \delta^0 - \sigma \circ \delta^1 + \sigma \circ \delta^2 \\ &= [v_1, v_2] - [v_0, v_2] + [v_0, v_1] \end{aligned}$$

Now applying ∂_1

$$\begin{aligned} (\partial_1 \circ \partial_2)(\sigma) &= \partial_1([v_1, v_2] - [v_0, v_2] + [v_0, v_1]) \\ &= \partial_1[v_1, v_2] - \partial_1[v_0, v_2] + \partial_1[v_0, v_1] \\ &= (v_2 - v_1) - (v_2 - v_0) + (v_1 - v_0) \\ &= v_2 - v_1 - v_2 + v_0 + v_1 - v_0 \\ &= 0 \end{aligned}$$

Note that this boundary property from **Theorem 1** implies that

$$(\partial_n \circ \partial_{n+1})(c) = 0, \, \forall c \in C_{n+1}(K)$$

$$\implies \partial_{n+1}(c) \in \ker \partial_n \text{ for every such } c$$

$$\implies \operatorname{im} \partial_{n+1} \subseteq \ker \partial_n$$

Definition 10. We define the *n-th simplicial homology group* as

$$H_n(K) := \ker \partial_n / \operatorname{im} \partial_{n+1}$$

Observe that since both $\ker \partial_n$ and $\operatorname{im} \partial_{n+1}$ are subgroups of $C_n(K)$, the quotient $H_n(K)$ is itself a finitely-generated abelian group.

Additionally, an element of $H_n(K)$ is the class

$$[x] = x + \operatorname{im} \partial_{n+1} = \{x + y : y \in \operatorname{im} \partial_{n+1}\}, \text{where } x \in \ker \partial_n$$

Definition 11. The *n-th Betti number* is given by

$$\beta_n(K) := \operatorname{rank} H_n(K)$$

2.2 Homology

Homology is a tool in algebraic topology that identifies and classifies the topological invariants of spaces. Given a chain complex (C_k, ∂_k), homology measures the *failure of exactness*: the extent at which cycles fail to be boundaries at each C_k. The key tools used are Betti numbers, where each one gives us a geometrical interpretation of the complex:

- β_0 counts the number of connected components (separate pieces) in the space.
- β_1 counts the number of 1-dimensional circular holes (like hole in donut or a loop in a graph).
- β_2 counts the number of 2-dimensional enclosed cavities (like the hollow interior of a sphere or balloon).

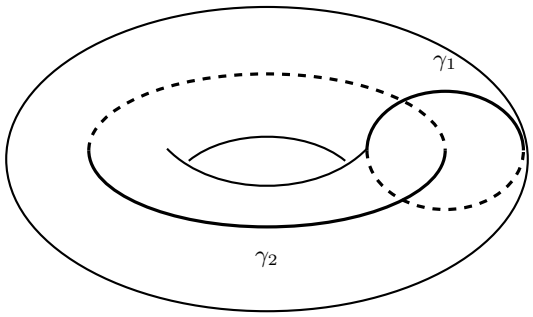


Figure 4: The torus has $\beta_1 = 2$. The two independent 1-cycles are the meridian γ_1, running around the tube, and the longitude γ_2, running around the central hole. Neither bounds a 2-chain, and neither is homologous to the other. Here $\beta_0 = 1$ (one component) and $\beta_2 = 1$ (one enclosed cavity).

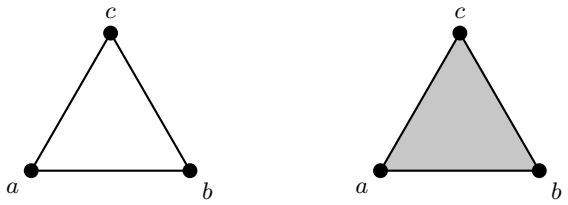


Figure 5: Left: the complex K containing only three vertices and three edges. The cycle $\{a,b\}-\{a,c\}+\{b,c\}$ generates $\ker \partial_1$ and bounds nothing, so $\beta_1 = 1$. Right: adjoining the 2-face $\{a,b,c\}$ makes that cycle a boundary, and β_1 drops to 0.

Everything above can be reproduced with coefficients in any field $\mathbb{F}$ instead of $\mathbb{Z}$: chains become $\mathbb{F}$-vector spaces with the simplices as a basis, boundary maps become linear, and $H_n(K;\mathbb{F})$ becomes an $\mathbb{F}$-vector space whose dimension we take as the Betti number. Over $\mathbb{Z}/2\mathbb{Z}$, the signs $(-1)^i$ disappear since $-1 \equiv 1$, and orientation no longer matters. Persistent homology requires a field [Section 2.3], and the software computes over $\mathbb{Z}/2\mathbb{Z}$, so from now on all homology is taken with coefficients in $\mathbb{Z}/2\mathbb{Z}$.

2.3 Persistent Homology

We are able to compute the topological invariants given to us by the Betti numbers through a process called *persistent homology*. This method consists of creating simplicial complexes from a point cloud through a filtration and computing homology on the built simplicial complex.

Starting from a point cloud $S = \{x_1, ..., x_n\} \subset \mathbb{R}^d$, there are two main types of filtration that one can compute: through a Vietoris-Rips complex, and through a Čech complex.

Given $\epsilon > 0$, the Čech complex is defined as follows:

$$\check{C}_\epsilon(S) = \{\tau \subseteq S \mid \bigcap_{x \in \tau} B_\epsilon(x) \neq \emptyset\}$$

So a set of points connect if all closed balls $\overline{B}_\epsilon(x_i)$ have a common point. By an important result in algebraic topology, the Nerve Theorem [6], this approximation is the "correct" one topologically. Concretely, the Čech complex is homotopy equivalent (i.e. they can be continuously deformed into one another through shrinking, expanding, or bending, without tearing) to the union of balls $\bigcup B_\epsilon(X_i)$.

On the other hand, the Vietoris Rips complex is given by

$$VR_\epsilon(S) = \{\tau \subseteq S \mid d(x, y) \leq 2\epsilon, \forall x, y \in \tau\}$$

Note that VR_ϵ depends on S only through its pairwise distances; no embedding is required, and (S, d) need only be a finite metric space. This is what makes VR available for data given as a distance matrix, whereas the Čech complex, which requires balls in an ambient space, is not.

Connections between two points trigger when their distance is less than or equal to 2ϵ. As only pairwise data is needed, it is computationally cheaper to compute, and storing information only requires a distance matrix. Nonetheless, it is not as topologically accurate as the Čech complex. Also, as they are interleaved

$$\check{C}_\epsilon(S) \subseteq VR_\epsilon(S) \subseteq \check{C}_{\sqrt{2}\epsilon}(S)$$

we prefer VR's computational efficiency for a "good enough" approximation.

Note that the complex computed by the VR is determined by one's choice of ϵ. If we increase the parameter until we have a fully connected graph, we get a nested sequence of complexes (a filtration):

$$X_{\epsilon_0} \subseteq X_{\epsilon_1} \subseteq ... \subseteq X_{\epsilon_m}$$

since increasing ϵ only adds simplices and does not remove them.

Now, for each complex X_ϵ, we build the chain groups $C_n(X_\epsilon, \mathbb{Z}/2\mathbb{Z})$ and boundary operators $\partial_n : C_n \to C_{n-1}$, giving us the homology groups $H_n(X_\epsilon)$ with Betti numbers $\beta_n(X_\epsilon)$ at each scale. This filtration transforms Betti numbers from single integers into a function of ϵ; a function that describes the dynamics of the topology as ϵ increases. Still, the function itself discards *which* invariants (holes) survived. For example, for $i \leq j$, if $\beta_1(X_{\epsilon_i}) = \beta_1(X_{\epsilon_j}) = 3$, we cannot tell if the same three holes persisted or whether two of them died and another two were born. In order to recover that, we need to compare the homology groups to each other, not just their dimensions.

The comparison comes as a result of the nesting $X_{\epsilon_i} \subseteq X_{\epsilon_j}$ as every n-face τ of X_{ϵ_i} is itself an n-face of X_{ϵ_j}. This fact allows us to define $\iota_n : C_n(X_{\epsilon_i}) \to C_n(X_{\epsilon_j})$ sending $\tau \mapsto \tau$, which is an inclusion that commutes with ∂_n:

$$\begin{array}{ccc} C_n(X_{\epsilon_i}) & \xrightarrow{\iota_n} & C_n(X_{\epsilon_j}) \\ \downarrow{\scriptstyle \partial_n} & & \downarrow{\scriptstyle \partial_n} \\ C_{n-1}(X_{\epsilon_i}) & \xrightarrow[\iota_{n-1}]{} & C_{n-1}(X_{\epsilon_j}) \end{array}$$

so that $\partial_n \circ \iota_n = \iota_{n-1} \circ \partial_n$, inducing a linear map on homology groups:

$$f_{i,j} : H_n(X_{\epsilon_i}) \to H_n(X_{\epsilon_j})$$

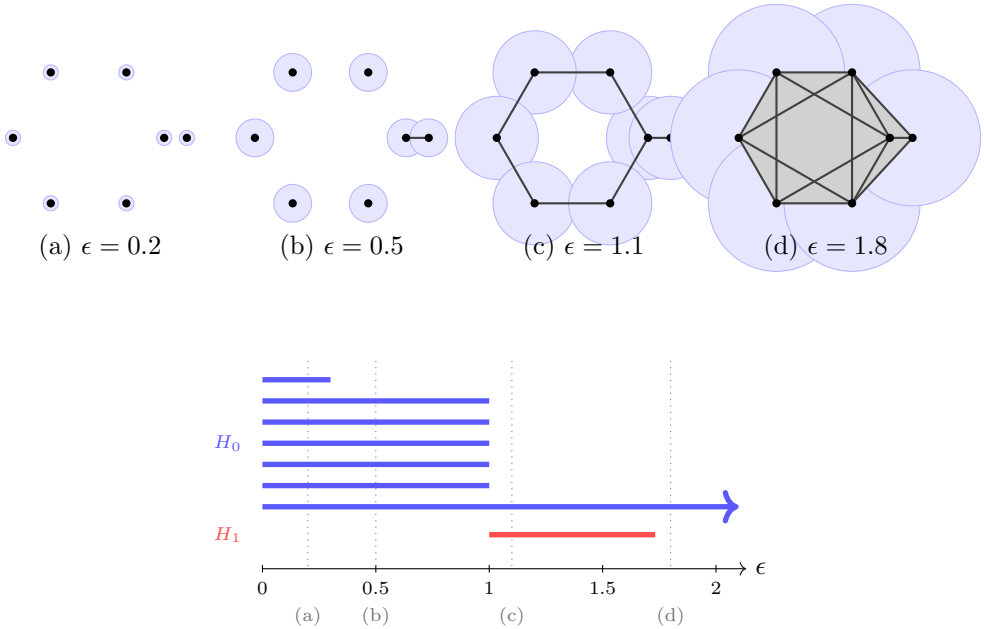


Figure 6: A Vietoris–Rips filtration on seven points and its barcode. Shaded discs have radius ϵ; an edge appears when two discs meet, i.e. when $d(x, y) \leq 2\epsilon$. (a) All points are isolated, $\beta_0 = 7$. (b) The close pair V_1, P joins at $\epsilon = 0.3$, killing one H_0 class; this short bar is the signature of a feature living at a single scale. (c) The ring closes at $\epsilon = 1$: $\beta_0 = 1$ and a single H_1 class is born. (d) By $\epsilon = \sqrt{3}$ enough triangles are present for the ring to bound, and the H_1 class dies. The five H_0 bars ending at $\epsilon = 1$ die simultaneously; by the elder rule the oldest class survives, giving the infinite bar.

Definition 12. A sequence of $\mathbb{Z}/2\mathbb{Z}$-vector spaces and linear maps

$$H_n(X_{\epsilon_0}) \xrightarrow{f_{0,1}} H_n(X_{\epsilon_1}) \xrightarrow{f_{1,2}} \cdots \xrightarrow{f_{m-1,m}} H_n(X_{\epsilon_m})$$

is called the *persistence module* of the filtration in degree n. For simplicity, let $M_i := H_n(X_{\epsilon_i})$ so we have

$$M_0 \xrightarrow{f_{0,1}} M_1 \xrightarrow{f_{1,2}} \cdots \xrightarrow{f_{m-1,m}} M_m$$

Definition 13. The *(i,j)-th persistent Betti number* is

$$\beta_n^{i,j} := \operatorname{rank} f_{i,j} = \dim_{\mathbb{Z}/2\mathbb{Z}}(\operatorname{im} f_{i,j})$$

It counts the classes born at or before ϵ_i that are still nonzero at ϵ_j. The ordinary Betti number is the diagonal case $\beta_n^{i,i} = \beta_n(X_{\epsilon_i})$.

A class $\alpha \in H_n(X_{\epsilon_i})$ is said to be **born** at ϵ_i if $\alpha \notin \operatorname{im} f_{i-1,i}$. Similarly, a class α **dies** entering ϵ_j if $f_{i,j-1}(\alpha) \neq 0$ but $f_{i,j}(\alpha) = 0$. This assigns an interval $[\epsilon_i, \epsilon_j)$ to each class, with $\epsilon_j = \infty$ if the class never dies. When two classes merge, by convention the younger one dies (the *elder rule*).

However, this chain of vector spaces is a complicated object to classify directly. For a better manipulation, we can fuse them into an algebraic object

$$\mathbb{M} := \bigoplus_{i=0}^{m} M_i$$

Now we can define an action of the polynomial ring $\mathbb{Z}/2\mathbb{Z}[t]$ on $\mathbb{M}$ by

$$t \cdot m := f_{i,i+1}(m) \text{ for } m \in M_i$$

and extending linearly, with t^k acting as k applications, i.e. $t^k \cdot m = f_{i,i+k}(m)$. So t is pushing – advancing – forward one step in the filtration.

Since $\mathbb{Z}/2\mathbb{Z}$ is a field, $\mathbb{Z}/2\mathbb{Z}[t]$ is a principal ideal domain (a commutative ring with multiplicative identity that has no zero divisors, in which every ideal can be generated by a single element). This lets us apply the classification of finitely generated modules over a PID.

Theorem 2 (Structure theorem). Over a field, a persistence module of finite type decomposes uniquely, up to isomorphism, as a direct sum of interval modules [7].

This theorem is what guarantees that the birth-death assignment above is well defined and preserves information: the intervals are a complete invariant of the module. In fact, $\beta_n^{i,j}$ is recovered as the number of intervals containing $[\epsilon_i, \epsilon_j]$.

Definition 14. The multiset of intervals in degree n produced by Theorem 2 is the **persistence barcode** $\mathcal{B}_n$. Recording each interval $[\epsilon_i, \epsilon_j)$ as a point (ϵ_i, ϵ_j) in the plane gives the **persistence diagram** D_n, together with the diagonal $\{\epsilon_i = \epsilon_j\}$ counted with infinite multiplicity.

The **persistence** of a feature is its lifespan $\epsilon_j - \epsilon_i$, i.e. its vertical distance from the diagonal. Pairs close to the diagonal represent features appearing and disappearing within a narrow range of scales, which is generally attributed to sampling noise, whereas pairs far from it indicate structures that are stable across a wide range of ϵ.

This multiset of birth-death pairs is the entire output of the computation: the algorithm takes a distance matrix and a maximum dimension and returns exactly this. Barcodes and persistence diagrams are two ways of plotting the same array. Both are used in Section 4 to summarize the topology of each recording.

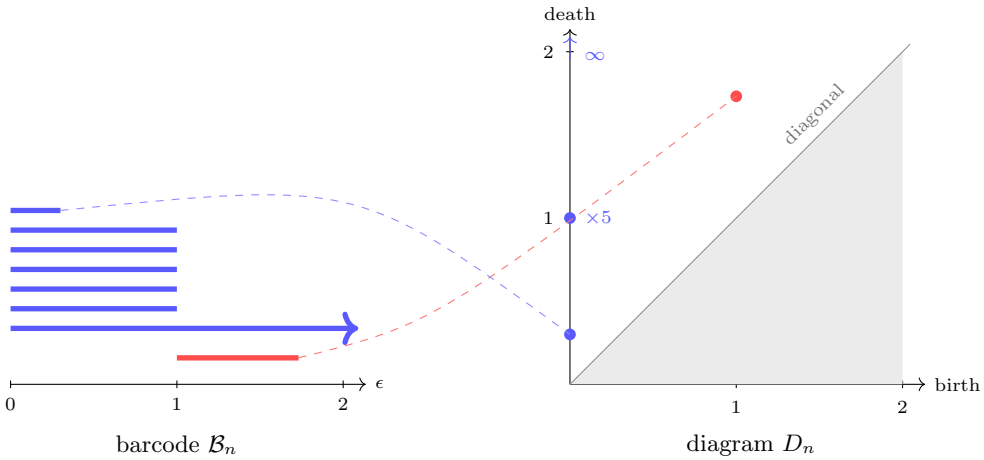


Figure 7: The same multiset of birth–death pairs rendered two ways. Each interval $[\epsilon_i, \epsilon_j)$ on the left is the point (ϵ_i, ϵ_j) on the right; the dashed guides mark two correspondences. Every H_0 class is born at $\epsilon = 0$, so H_0 points lie on the vertical axis. The vertical distance from the diagonal is the persistence $\epsilon_j - \epsilon_i$: the short H_0 bar sits close to the diagonal, the long H_1 bar far from it. Both serve as different displays for the same array.

2.4 Topological Data Analysis

Topological Data Analysis states that data has a shape, and that shape carries valuable information. It is often framed under the assumption that the data point-cloud is a finite sample from an underlying continuous geometric space or manifold. In this way, a simplicial complex is built from the point cloud to approximate the underlying manifold and thus study the properties of the family of complexes obtained. In this paper, no such manifold or ambient space is assumed: the electrodes form a dissimilarity space, and the filtration is built directly from the dissimilarity matrix [Section 4.2].

3 Data

We used the EEGMAT dataset (PhysioNet, eegmat/1.0.0) of 36 subjects; each recorded at rest with their eyes closed ($\sim$ 182s) and while performing serial subtractions of a two-digit number from a four-digit number ($\sim$ 62s). The EEG setup consists of 19 standard scalp electrodes positioned in the 10-20 montage. Signals are cut into four-second, non overlapping intervals of 640 samples. As the rest recording is around three times longer than the task recording, we take the first fifteen windows of each recording; that is, the first minute of each recording. This leaves us with a total of 36 subjects $\times$ 15 windows per class $\times$ 2 classes per subject = 1080 windows.

4 Methods

4.1 Connectivity

Each window is band-pass filtered twice by a zero phase, fourth-order Butterworth filter, once into alpha (8-13 Hz) and once into beta (13-30 Hz). After that, the analytic signal is obtained through a Hilbert transform, allowing us to extract hidden information from a signal, such as instantaneous frequency, amplitude, and phase at any exact moment in time. The Hilbert transform H shifts every frequency component of the filtered signal $x(t)$ by 90 degrees. Combining the two gives the complex analytical signal $z(t) = x(t) + iH(x) = A(t)e^{i\phi(t)}$, whose modulus $A(t) = |z(t)|$ is the instantaneous amplitude (used by AEC) and whose argument $\phi(t) = \arg z(t)$ is the instantaneous phase (used by wPLI).

Explicitly, wPLI and AEC are computed by

$$\mathrm{wPLI}_{ij} = \frac{\left|\sum_{t=1}^{L} \mathrm{Im}\left(z_i(t)\,\overline{z_j(t)}\right)\right|}{\sum_{t=1}^{L} \left|\mathrm{Im}\left(z_i(t)\,\overline{z_j(t)}\right)\right|} \in [0,1], \qquad \mathrm{AEC}_{ij} = \left|\mathrm{corr}_t(|z_i(t)|,\, |z_j(t)|)\right|.$$

Here, the sums and the correlation run over the 640 time samples of a single window, rather than over trials as in the original definition [8]. For both measures, the diagonal entries of the connectivity matrix are set to $M_{ii} = 1$, so that $D = 1 - M$ satisfies $D_{ii} = 0$ [Figure 8].

Notice that for each combination (alpha-wPLI, alpha-AEC, beta-wPLI, beta-AEC) we have a 640-dimensional characteristic vector for each electrode (from each sample). But as we are interested in making pair-wise comparisons from the electrodes, we can relate their values from their 640 samples by "assigning" a value for each pair, yielding a total of four 19×19 symmetric connectivity matrices per window.

4.2 From coupling to a filtration

The 19 electrodes are abstract vertices of a *finite dissimilarity space* (a finite set X with a function $d : X \times X \mapsto \mathbb{R}$ such that $d(x, y) = d(y, x)$ and $d(x, x) = 0$; unlike a metric, it need not satisfy the triangle inequality), not points in $\mathbb{R}^d$. No embedding is computed in any stage, and the filtration is directly taken from the 19×19 "distance" matrix. This enables the descriptor to be invariant to the labeling of the electrodes, as

it only concerns itself with the mutual distances. Each connectivity matrix M is converted to a dissimilarity $D = 1 - M$ so that strongly coupled electrodes lie close together. Note that D is symmetric and non-negative with zero diagonal, but it is not a metric. That, regardless, is not relevant, as the Vietoris-Rips construction needs only symmetry and non-negativity, so the filtration and the structure theorem apply unchanged. And while the Čech construction and interleaving with V-R do not hold here, we do not rely on it for the computation.

The software uses the diameter convention, so its threshold ϵ corresponds to 2ϵ in Section 2.3. A set of electrodes enters the V-R complex at threshold ϵ when every pair within has $D \leq \epsilon$. A simplex therefore appears at the largest of its pairwise dissimilarities. Increasing the value of ϵ adds more simplices, which gives a nested filtration; homology is then computed over $\mathbb{Z}/2\mathbb{Z}$. On 19 vertices, as H_2 classes are rare and short-lived, they are omitted. Every H_0 class is born at $\epsilon = 0$, so only its death time is informative [Figure 8].

The four matrices [Section 4.1] are treated as four separate spaces. Each is filtered independently and yields its own pair of diagrams; they only meet when their vectorizations are concatenated.

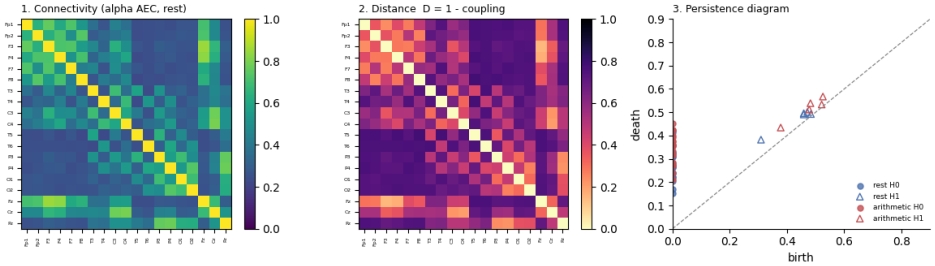


Figure 8: **1.** One of the four computed matrices, alpha-AEC, which shows the electrodes' coupling strength with each other. **2.** The only thing that changes is the values are inverted, now the [0,1] scale represents how far each pair of electrodes are: strong coupling becomes short distance. The new values are computed by Distance = 1 - Coupling, so that the V-R can connect points that sit close together first. **3.** Circles H_0 are moments when two clusters of electrodes merge; every electrode starts at its own piece at $\epsilon = 0$, which is why all births are 0. But we can notice that red circles (task) reach higher, so coupling is weaker overall during the task. Triangles H_1 are loops, and their distance from the diagonal is their lifespan. Note that they sit close to the diagonal, so they close almost as soon as they open, which is to be expected at 19 points. This last panel was computed from the state-average matrix for illustration; the classifier uses per-window diagrams.

4.3 Vectorization

Recall that a persistence diagram is a list of birth-death pairs. The count of H_1 features varies across windows, and across windows this count ranges from 0 to around 50; so to standardize them, we must map them to a fixed-length vector before feeding them to the classifier. Each block's diagrams are summarized in four ways:

- H_0 death histogram (12 numbers). How many merges of pieces (β_0) happen in each of 12 slices of the threshold range.
- Persistence image of the loops (100 numbers). A 10×10 unit-grid of pixels 0.1 wide, where each loop becomes one point (x, y) s.t. x = when it appears (birth), y = how long it lasts (death-birth). Each point is then replaced by a Gaussian centered on it, multiplied by a weight that grows linearly with its persistence, so longer-lived loops count more. Then, each pixel's value is the integral of the resulting surface over that pixel.
- Betti numbers (40 numbers). β_0 and β_1 were computed individually at 20 different threshold values.

- Persistence moments (16 numbers). 8 summary statistics for the pieces (β_0) and the loops (β_1): number of features, their total, longest, average, and median lifetimes, how evenly their lifetimes are spread (entropy), and average birth and death times.

By construction, these are fixed-size, which gives us $12+100+40+16 = 168$ numbers per block. And with four blocks, this leaves us with a 672-dimensional vector per window, which serves as the final topological descriptor.

Although coherent, we have to be careful with this build. We can notice that if we shift every coupling value by a constant, the overall topology should not be altered; however, because the bins are rigid and do not move, this shift would cause the features to land in different pixels, making the classifier see different numbers. To fix this, we center each of the four connectivity blocks on their average over the whole dataset before computing topology. This way, the overall coupling level always sits in the same place relative to the bins.

4.4 Classification and Validation

Features are standardized per dimension and classified by a support vector machine with a radial basis function kernel, $e^{-\gamma\|\mathbf{x}-\mathbf{x}'\|_2^2}$, which measures similarity between two points based on their distance to each other [9]. The SVM was trained using cross-validation, five times, once per outer fold. The 36 subjects were split into four groups of 7 and one of 8, with each group taking one turn as the test set; these rounds produce the reported accuracy. The remaining ~ 29 subjects were split again into 3 groups, which were used as the validation set to choose the parameters. Every subject was tested exactly once by a model that never saw their data [Figure 9]. We are then left with 5 SVMs, each having tested ~ 7 subjects with joint labels for all 1080 windows.

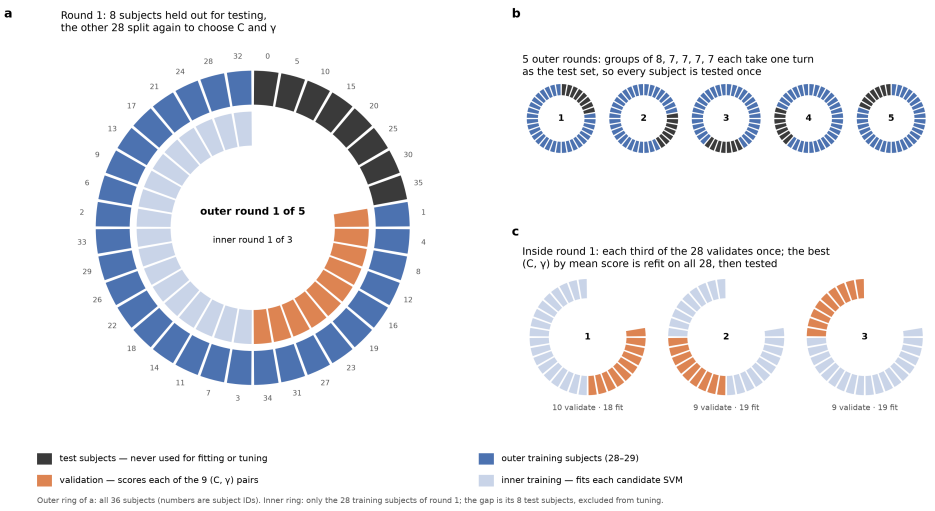


Figure 9: Cross-validation on all 36 subjects.

4.5 Normalization and pooling

Due to the influence of external factors at the time of recording, skull thickness differences, and individual variability, a single four-second window is a noisy observation of the data. So before classification, each feature was z-scored $x' = \frac{x-\mu_S}{\sigma_S}$, where μ_S and σ_S are the mean and standard deviation, respectively, of all 30 windows of subject S. The normalization is applied to every subject, including those held out for testing, using that subject's own windows. Although no labels are used, this procedure implicitly relies on each subject contributing equal numbers of rest and task windows. In a deployed setting, this assumption may not be available since the proportion of each state in a new subject's data is unknown. The reported accuracies should then be read as performance given a balanced, unlabeled calibration session per subject.

After classification, the classifier labels each four-second window separately. The 15 windows of a single recording are then combined by majority vote into a single label for that recording; note that, as 15 is odd, voting cannot result in a tie. Then, each subject receives either 0, 1/2, or 1, and the mean of the 36 scores is the reported accuracy.

4.6 Matched baselines

Two baselines are computed from the same windows and pushed through the identical classifier so only representation differs. Band-power is the field-standard feature: the log-variance of each channel in each band, taken from the filtered signal before the Hilbert transform. Raw connectivity is the sharper control, it takes the pairwise coupling values as input, but keeps them as raw numbers instead of computing their shape. This means that any difference between it and the topological descriptor is attributable to the descriptor alone [Figure 10].

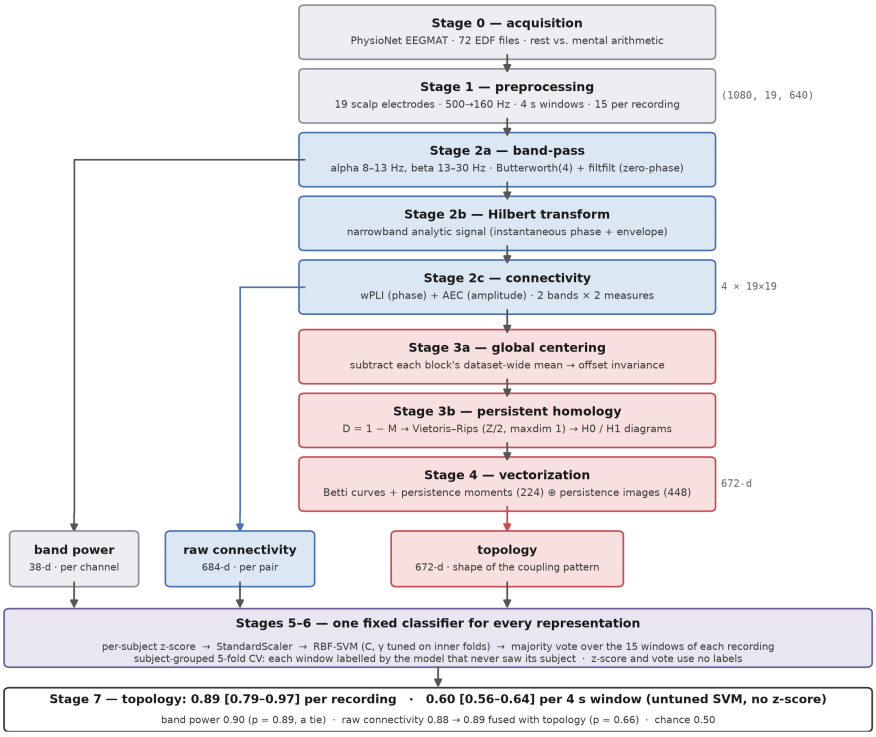


Figure 10: Pipeline architecture diagram

5 Results

The topological classifier reaches 0.89 per-recording accuracy (95% bootstrap CI 0.79-0.97). Band power reaches 0.90, and raw connectivity reaches 0.88. It is important to note that, per window, without per-subject normalization or tuning, the topological features give 0.60 (95% CI 0.56-0.64) across 1080 decisions. The three representations are statistically indistinguishable. The combination of raw connectivity and topology adds up to 0.89, which improved three subjects and harmed two [Figure 11].

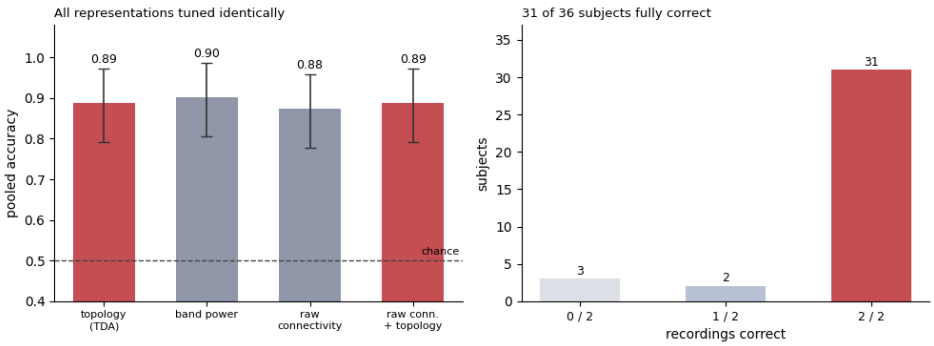


Figure 11: Left: per recording accuracy of each representation, all tuned identically. Right: how many subjects have zero, one or both recordings correctly classified.

6 Discussion

6.1 Shape is sufficient representation of cognitive state

The features were used to represent the shape of each coupling space: how its components merge and what loops form. As they contain no information regarding the labeling of the electrodes, permuting them leaves the persistence diagram exactly the same. The fact that this representation classifies the two states at 0.89 per recording shows that the geometry of coordinated cortical activity is sufficient to identify a cognitive state without the need for spatial location. Strictly speaking, topology is a reformulation of the coupling values since it is computed from them.

6.2 Relation to previously reported accuracies

Most results come from large within-subject splits, where the same subject appears in both training and testing [10, 11]. Other classification goals were involved, such as seizure detection [12], which represents a more significant contrast than a cognitive shift. Additionally, TDA-extracted features are usually fed to more complex classifiers like CNNs [12, 13], instead of a kernel classifier alone.

6.3 Limitations

- The difference in duration between both states is a factor to take into account: as the rest state lasts longer, the more informative activity may lie in the later part of the recording, while only the first 60 seconds were analyzed.
- The states differ in more than cognition; as the task involved major sustained effort, they might include subvocalization or eye movement, which may play a significant role in altering the recording and thus the classification; therefore, we may not be classifying mere brain activity.
- The frequency bands, alpha and beta, were fixed a priori; theta (4-8 Hz), which is associated with arithmetic and working memory, was deliberately discarded and not taken into account. The reason was to avoid having to deal with an eye-contamination verification process of the recording, as the contamination risk usually concentrates in low frequencies and at the frontal electrodes, where frontal-midline theta is measured. Also, as EEGMAT does not have electrooculography (EOG), standard EOG regression is not available, so we would have to rely on the preprocessed ICA that the data source offers. Nonetheless, leaving theta out does not weaken the main claim: the question was whether topology can tell two states apart, so showing that with two less-contaminated bands is a cleaner test.
- As in all neuroscience, generalization is not immediate: we are working with a relatively small dataset consisting of two high-contrast states. Generalizing the topological claim could get harder as the sample grows and different brain states get involved.

- The per-subject normalization, as stated in Section 4.5, uses all of each subject's windows and assumes a balanced number of rest and task windows.
- All results come from a single subject-grouped 5-fold split. Since the differences between representations amount to one or two subjects at most, no representation should be considered superior.

7 Conclusion

Persistent homology of EEG functional connectivity distinguishes rest from mental arithmetic at 89% accuracy per recording and 60% per four-second window across unseen subjects. A very important point is that the descriptors we used are interpretable: counts of connected components, H_0, and loops, H_1, across different coupling scales. The shape of coordinated brain activity is sufficient to read out cognitive states.

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